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Increasing gonadorelin hydrochloride dose at the first GnRH of breeding Ovsynch enhances luteinizing hormone release, ovulatory response, and pregnancy in first-service lactating dairy cows

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ABSTRACT

The objective of this study was to evaluate the effect of administering 200 µg versus 100 µg of GnRH as gonadorelin hydrochloride (Factrel; Zoetis, Parsippany, NJ) at the first GnRH injection (G1) of the breeding Ovsynch in a Double-Ovsynch (DO) protocol on pregnancy per artificial insemination (P/AI) at first service in lactating dairy cows. The study was conducted across 4 commercial dairies located in California (n = 2) and Wisconsin (n = 2). We also assessed the effect of GnRH dose on ovulatory response and LH release. Each week, cows were blocked in a 2 × 2 factorial design by parity (primiparous or multiparous) and GnRH dose (100 µg or 200 µg) within each herd. The primary outcome was P/AI following estrus synchronization and timed AI. Cows were excluded if they were inseminated outside the synchronization window, culled before pregnancy diagnosis, or missed scheduled GnRH or PGF_{2α} treatments. Dinoprost tromethamine (12.5 mg/mL; Lutalyse HighCon, Zoetis) served as the PGF_{2α} source. A total of 6,621 cows met the inclusion criteria (2,525 primiparous; 4,096 multiparous). Of these, 3,310 received 100 µg GnRH (2 mL), and 3,311 received 200 µg GnRH (4 mL). A subset of 505 cows from herd A underwent transrectal ultrasonography to assess ovulatory response at G1 and 7 d later (first PGF_{2α} of DO), with concurrent serum progesterone (P4) sampling. An additional subset of 18 cows had blood samples collected at 0 (pretreatment) and at 30, 60, 90, 120, 150, 180, 210, 240, 300, and 360 min post-treatment to measure LH concentrations. The final statistical model for P/AI included fixed effects of treatment, parity, and their interaction, with random effects

of herd, service sire, herd-by-treatment-by-parity, and month fresh nested within herd-by-parity. Treatment and parity significantly affected P/AI, although their interaction was not significant. Least squares means for P/AI were higher with 200 µg than 100 µg GnRH (51.8% vs. 48.6%). When stratified, the effect in primiparous cows was minimal (52.8% vs. 51.8%), but substantial in multiparous cows (50.7% vs. 45.4%). Cows treated with 200 µg GnRH had a greater ovulatory response than those receiving 100 µg (83.0% vs. 70.8%). Parity also influenced ovulation, though no treatment-by-parity interaction was observed. When P4 was categorized (Low <3.0 ng/mL, High ≥3.0 ng/mL), a 3-way interaction emerged: in primiparous cows with Low-P4, ovulation was greater with 200 µg; in multiparous cows with High-P4, ovulation was greater with 200 µg. No treatment effect was observed on P4 concentrations 7 d post-treatment. Cows receiving 200 µg GnRH had higher peak LH and area under the curve than those receiving 100 µg, though peak timing did not differ. In summary, increasing the GnRH dose to 200 µg at G1 of the breeding Ovsynch in a DO program improved LH release, ovulatory response, and first-service P/AI, with the greatest benefit in multiparous cows.

Key words: GnRH, luteinizing hormone, first insemination, fertility

INTRODUCTION

The independent discovery of GnRH by Andrew Schally and Roger Guillemin in 1971 (Guillemin et al., 1971; Schally et al., 1971) was a landmark moment in reproductive biology. Their work, which earned the Nobel Prize in Physiology or Medicine in 1977, fundamentally advanced our understanding of neuroendocrine regulation and laid the foundation for therapeutic strategies across species. In dairy cows, early research on GnRH

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focused on dose response for LH and FSH (Schams et al., 1974) and its use for treating follicular cysts (Seguin et al., 1976). From the 1970s to the 1990s, numerous studies confirmed its efficacy for cyst treatment and improving pregnancy outcomes, as revised by Mee et al. (1990).

In the 1990s and early 2000s, GnRH became one of the cornerstones for the development of timed artificial insemination (AI) programs, such as Ovsynch. These programs revolutionized reproductive management by substantially improving submission rates and pregnancy per AI in dairy herds (Pursley et al., 1995, 1997a,b) and becoming indispensable tools for optimizing pregnancy outcomes and farm profitability (Giordano et al., 2011; Chebel et al., 2025). In the last 25 yr, extensive research has sought to refine the physiological understanding and application of these synchronization programs (Vasconcelos et al., 1999; Moreira et al., 2001; Bello et al., 2006), leading to the development of what today is referred to as fertility programs (Fricke and Wiltbank, 2022).

A quintessential example of a fertility program is the Double-Ovsynch (DO), which was developed to aid anovular cows and to increase the odds of having cows starting the Ovsynch (for breeding) cycle and in a favorable stage of the estrous cycle (i.e., d 6 to 7) to respond to the first GnRH of Ovsynch (Souza et al., 2008). At this stage, progesterone concentrations are lower and less likely to block GnRH-induced ovulation in the first-wave dominant follicle (Giordano et al., 2012). Indeed, ovulation, at the beginning of the Ovsynch, was characterized as the hallmark of its success, being associated with improved luteolysis, ovulation at second GnRH, and pregnancy per AI (P/AI; Pursley et al., 1995, 1997a,b).

The DO successfully aided anovular cows, increasing the percentage of cows cycling when compared with the classic Presynch-Ovsynch (Souza et al., 2008) and P/AI (Souza et al., 2008; Giordano et al., 2013; Carvalho et al., 2018). Moreover, other studies indicated and suggested several positive physiological responses of the DO, including optimized progesterone concentrations during follicular development (Giordano et al., 2013; Carvalho et al., 2015), which in turn has been proposed to create an optimum environment for follicle development and oocyte quality that blocks prolonged exposure to high LH pulses that presumptively prevents early meiotic resumption, reduces double ovulation, twin pregnancies, and pregnancy loss, and leads to ovulation of smaller follicles similar to heifers thought to be more fertile at timed AI (Giordano et al., 2013; Carvalho et al., 2018; Martins et al., 2018; Fricke and Wiltbank, 2022).

Yet, despite these advantages, the ovulatory response to the first GnRH in the breeding Ovsynch of the DO program remains suboptimal. Ovulation rates of ~72% are not markedly better than those observed with PGF_{2α}-based presynchronization (Souza et al., 2008). A likely

explanation is the high circulating progesterone (>3 ng/mL) at the time of the first GnRH dose of breeding Ovsynch of DO, driven by the presence of a mature corpus luteum (Souza et al., 2008; Giordano et al., 2013).

This hypothesis was evaluated by Giordano et al. (2012), who demonstrated that high progesterone at the time of GnRH treatment attenuates LH release. Importantly, increasing the GnRH dose from 100 to 200 µg elevated LH secretion regardless of progesterone level, suggesting a potential strategy to enhance ovulation. Building on this conceptual framework, Valdes-Arciniega et al. (2023) reported that 200 µg of GnRH as gonadorelin hydrochloride improved both ovulation and P/AI, although in primiparous cows, the benefits on P/AI were modulated by semen type. Similar findings in heifers, who also experience high progesterone inhibition of LH release at the synchronization onset (Lima et al., 2020), support this approach. Studies in heifers showed consistently positive results for ovulation (Colazo et al., 2023; Melo et al., 2024) and positive results for P/AI in one study (Colazo et al., 2023), with no difference in P/AI in another study (Melo et al., 2024).

Taken together, these findings suggest that a higher GnRH dose may consistently improve ovulation. However, its effect on LH release dynamics remains insufficiently characterized. Moreover, given the heterogeneous nature of P/AI in dairy cows (influenced by factors such as management, genetics, breed, parity, farm, season, and semen type; Lima et al., 2020), robust, randomized, multifarm studies are needed to understand better and validate this intervention.

The primary objective of this study was to compare the effects of administering 100 µg versus 200 µg of GnRH as gonadorelin hydrochloride (Factrel; Zoetis, Parsippany, NJ) at the first GnRH of the breeding Ovsynch in a DO program on P/AI at first service in lactating dairy cows across 4 commercial dairies. Secondary objectives included evaluating differences in ovulatory response and LH release between treatments. We hypothesized that cows receiving 200 µg of GnRH would exhibit higher LH release, ovulatory response, and P/AI than herdmates treated with 100 µg of GnRH.

MATERIALS AND METHODS

Ethical Approval

The study component for P/AI was conducted in accordance with 7 USC 54 at 4 commercial dairies operated under routine management practices supervised by herd veterinarians. The experimental procedures carried out in the subset of cows receiving ultrasonography examination and blood sampling for hormonal measurements received an ethical review by the Institutional Care and

Use Committee of the University of California, Davis (protocol no. 22792).

Cows, Diets, Housing, and Management

This study was conducted from November 2020 to October 2021 across 4 commercial dairies (2 located in California and 2 in Wisconsin) that agreed to participate due to their consistent use of DO-based estrus synchronization protocols. Each farm followed standard commercial practices. Cows were housed in freestall barns or open dry lots with shade structures and were managed using headlock systems for handling and treatment. Three dairies exclusively housed Holstein cows, while one herd consisted of 91% Holstein and 9% Jersey or Jersey-Holstein crossbred cows. All herds TMR formulated to meet or exceed NRC (2001) nutritional requirements and is composed of corn silage, alfalfa hay, soybean meal-based concentrate, and locally available byproducts (e.g., distillers' grains, cottonseed, almond hulls), supplemented with minerals and vitamins. Herd management data were maintained using DairyComp 305 (Valley Agricultural Software, Tulare, CA) or BoviSync (BoviSync, Fond du Lac, WI). Table 1 provides herd-specific demographic and management details, including herd size, voluntary waiting period, semen usage, DO program timing, and pregnancy check protocols.

Experimental Design

Within each herd, cows were ranked in ascending order of genomic prediction of daughter pregnancy rate GDPR to balance predicted fertility across treatment groups, and a randomized complete block design with a 2×2 factorial arrangement was implemented, considering 2 levels

of parity (primiparous and multiparous) and 2 doses of GnRH (100 or 200 μ g; Factrel, Zoetis). Each cow served as the experimental unit. Cows initiating the DO protocol on the same day were grouped into weekly cohorts and randomized by parity into treatment groups, ensuring a balanced distribution within each cohort.

The DO protocol began with a GnRH injection, followed 7 d later by PGF 2α , and 3 d later by a second GnRH injection, concluding the presynchronization phase. Seven days after the last presynch GnRH, cows received the first GnRH (G1) of the breeding Ovsynch and were administered either 100 μ g (2 mL) or 200 μ g (4 mL) of GnRH (Figure 1). This was followed 7 and 8 d later by PGF 2α treatments (25 mg/2 mL, Lutalyse HighCon, Zoetis) and a second GnRH dose (~32 h later), followed ~16 h later by timed AI (Figure 1).

All inseminations were performed by routine herd personnel using sires selected as part of the herd-specific reproductive management program. Pregnancy diagnoses were performed by transrectal palpation or ultrasonography according to herd-specific schedules (32–39 d post-AI for the first diagnosis and 66–67 d post-AI for the recheck). For each insemination, pregnancy P/AI was calculated separately at the first pregnancy diagnosis and again at the recheck. Pregnancy loss was defined as the proportion of cows diagnosed pregnant at the first examination but determined nonpregnant at the recheck.

To be included in the final analysis, cows had to meet the following criteria: complete record of identification, breed, parity, calving date, injection schedule, AI date, day of the week AI occurred, and pregnancy diagnosis at 32 to 39 d post-AI. Cows that received fewer than 7 hormonal injections (GnRH and PGF 2α), missed scheduled treatments, or were inseminated outside the scheduled day due to estrus detection were excluded. The hormonal

Table 1. Detailed demographics and herd characteristics for farms enrolled in the current study

Item	Herd A	Herd B	Herd C	Herd D
Farm location	California	California	Wisconsin	Wisconsin
Herd size, n	5,600	2,200	9,500	4,400
Herd breed distribution	100% Holstein	91% Holstein, 9% Jersey or Jersey crossbred	100% Holstein	100% Holstein
Average 305 d mature equivalent milk yield, kg	12,757	11,713	14,528	14,982
Start of estrus synchronization, DIM $\pm$ 3 d	61	61	Primiparous: 46 Multiparous: 38	43
First insemination, DIM $\pm$ 3 d	88	88	Primiparous: 73 Multiparous: 65	70
Semen types used	Angus, sexed Holstein	Angus, Charolais, sexed Holstein	Angus, Simmental, sexed Holstein, sexed Jersey	Limousin, conventional Holstein
Targeted days postinsemination at first pregnancy exam	32	35	39	32
Targeted days postinsemination at second pregnancy exam	67	66	67	67
Pregnancy diagnosis method used	Ultrasound	Ultrasound	Transrectal palpation	Ultrasound

Enrollment Source and Timeline of Activities and Sampling

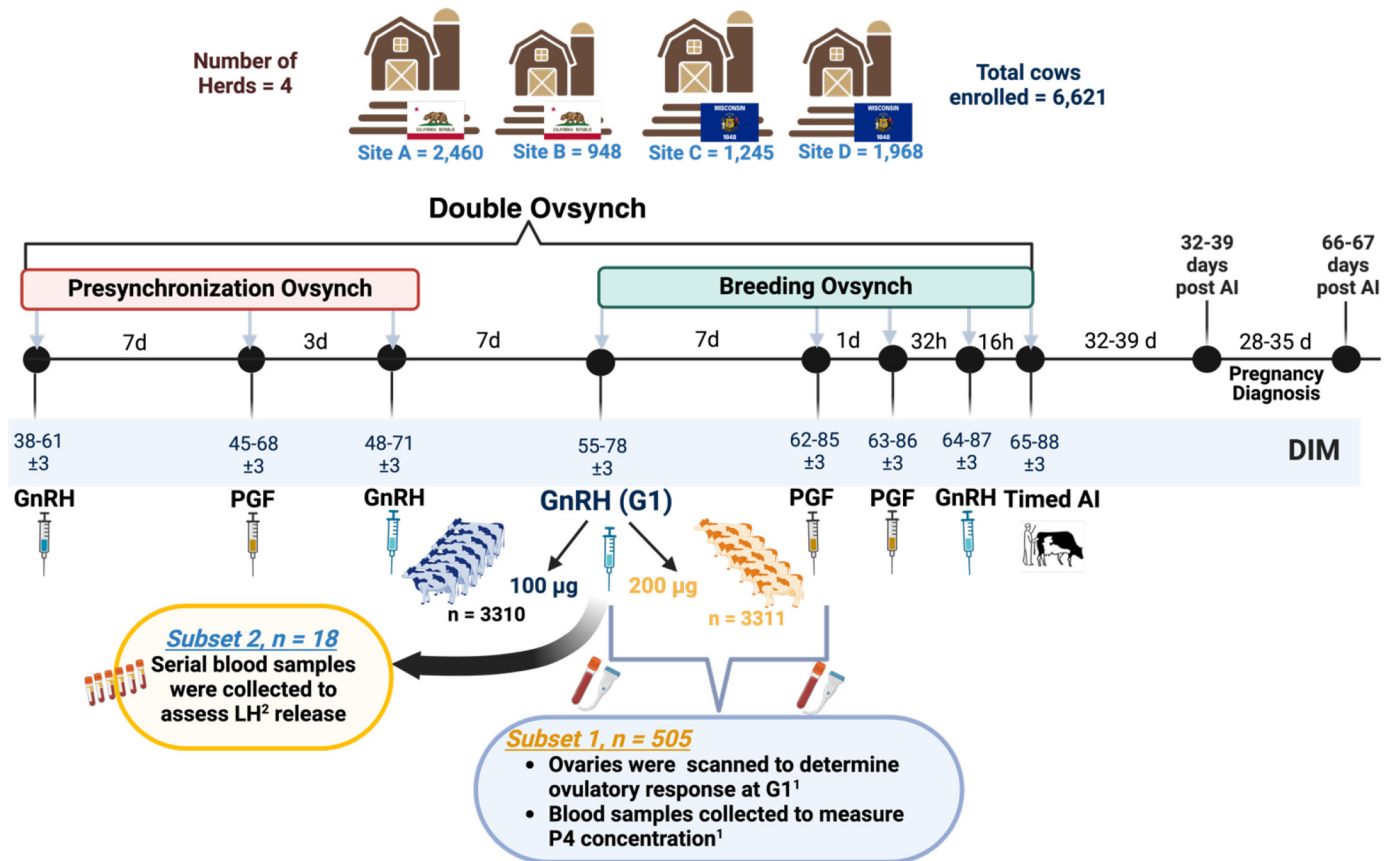


Figure 1. Detailed depiction of herd and cow enrollment sources, reproductive management activities, experimental design allocation, and subset sampling used in the study. ¹A subset of 505 cows had their ovaries at the time of the first GnRH (G1) of the breeding Ovsynch of the Double-Ovsynch and again 7 d later at the PGF_{2α} treatment to characterize ovarian structures present and determine ovulatory response, while concurrently blood samples were collected to measure the blood concentration of progesterone. ²A second subset of 18 cows had serial blood samples collected at 0 (before G1) and 30, 60, 90, 120, 150, 180, 210, 240, 300, and 360 min after treatment to measure the serum concentration of LH.

treatment sequence was GnRH (−27 d), PGF_{2α} (−20 d), GnRH (−17 d), GnRH (−10 d), PGF_{2α} (−3 d), PGF_{2α} (−2 d), and GnRH (−1 d).

Ovarian Ultrasound and Ovulatory Response Assessment

Each week, ~40 cows from herd A were randomly selected for ovarian ultrasound at G1 and again 7 d later at the first PGF_{2α} of the breeding Ovsynch. A total of 505 cows (345 primiparous, 160 multiparous) were evaluated (246 received 100 µg; 259 received 200 µg). Ultrasound examinations were performed using the Easi-Scan Go Veterinary Ultrasound (IMV Imaging, Rochester, MN). Ovarian structures were evaluated for the presence of follicles ≥10 mm and corpora lutea (CL) ≥14 mm.

Ovulation was defined as the disappearance of a follicle ≥10 mm observed at G1 with a newly formed CL present

at the exact location 7 d later. Cows with no qualifying follicle at G1 but a new CL at PGF_{2α} were considered to have ovulated before G1. Follicle and CL diameters were estimated using on-screen gridlines; the volume of follicles and CL were calculated as the equation $4.1888 \times (\text{diameter of follicle or CL}/2)^3$, as previously described (Mercé et al., 2006). For CL with fluid cavities, cavity volume was subtracted from total CL volume to yield functional volume.

Blood Sampling and Hormone Assays

From the same subset of 505 cows examined by ultrasound, blood samples were collected at G1 and again 7 d later. Samples were drawn via coccygeal venipuncture into EDTA-containing evacuated tubes (Becton Dickinson, Franklin Lakes, NJ), chilled immediately, centrifuged at $2,000 \times g$ for 15 min at 5°C, and stored at

–80°C. Plasma progesterone (**P4**) was measured using a validated enzyme immunoassay (Munro and Stabenfeldt, 1984) with methodology adapted from Gomez et al. (2018) and Swanepoel et al. (2020). The assay sensitivity was 1.17 ng/mL, with intra- and interassay CV of 9.2% and 9.8%, respectively. Cows were categorized as **Low-P4** (≤ 3.0 ng/mL) or **High-P4** (> 3.0 ng/mL) to help assess the potential negative effect of High-P4 on ovulation and the subsequent benefits of a higher dose of GnRH, as suggested by previous studies (Giordano et al., 2012; Valdes-Arciniega et al., 2023).

To assess LH response, 18 cows (6 multiparous, 12 primiparous) with a CL > 16 mm and a follicle > 10 mm at G1 were randomly selected and assigned to receive either 100 μ g ($n = 10$) or 200 μ g ($n = 8$) GnRH. Blood samples were collected at 0 (pretreatment), 30, 60, 90, 120, 150, 180, 210, 240, 300, and 360 min postinjection. Samples were processed and stored as described above. The LH concentrations were quantified using an RIA validated for bovine serum (Bolt and Rollins, 1983; Ginther et al., 1999). Intra-assay CV were 19.2% (low control) and 12.3% (high control); interassay CV were 7.4% and 10.6%, respectively.

Statistical Analysis

The sample size for this study was based on detecting a 4-percentage-point difference in first-service P/AI between cows treated with 100 μ g versus 200 μ g of GnRH, as suggested by prior studies (Valdes-Arciniega et al., 2023) evaluating gonadorelin dose at the beginning of the breeding Ovsynch of the DO. Assuming a baseline P/AI of 48%, a 2-sided test with 80% power and $\alpha = 0.05$ required $\sim 1,508$ cows per treatment group. To enhance statistical power and ensure robustness across varying herd-level conditions, we enrolled 6,621 cows (3,311 and 3,310 per group) across 4 commercial dairy farms located in California and Wisconsin. The multisite design increased external validity by capturing geographic, environmental, and management variation. Potential clustering of outcomes within herds was accounted for by including herd and herd-by-treatment-by-parity as random effects in all statistical models, thereby preserving valid inference within this hierarchical study design.

Data analyses were performed using SAS/STAT 15.1 (SAS 9.4; SAS Institute Inc., Cary, NC). Binary outcomes (e.g., P/AI, ovulation, CL presence) were analyzed using logistic regression with the GLIMMIX procedure. The model for P/AI included fixed effects of parity, treatment, and their interaction, and random effects of herd, herd $\times$ treatment $\times$ parity, service sire, and month fresh nested within herd $\times$ parity. Least squares means and SEM were back-transformed and reported. In preliminary sensitivity models, GDPR (and its interactions)

did not affect outcomes; to preserve parsimony, GDPR was not retained in final models. Models with weekly cohorts nested within farms were also assessed, but because monthly calving nested by farm results in similar estimates and accounts for potential calving-related issues, management changes, and nutritional changes, and prevents overparametrization, it was the temporal factor included in our final model.

Continuous outcomes (e.g., P4 concentration, follicle/CL size, and volume) were analyzed using the GLIMMIX procedure with a Gaussian distribution. Count data (e.g., number of follicles or CL) were modeled using the GLIMMIX procedure using the Poisson distribution. For P4, cows were classified (High > 3.0 ng/mL or Low ≤ 3.0 ng/mL) categorically. Additional ovulation models included treatment, parity, P4 category, and their interactions.

The normality of LH measures (peak, mean, time to peak, area under the curve [AUC]) was assessed with PROC UNIVARIATE using Shapiro–Wilk and graphical diagnostics. The LH-related variables were analyzed by ANOVA using PROC MIXED. The area under the curve was calculated using the trapezoidal method. Significance was declared at $P < 0.05$, with trends reported for $0.05 \leq P < 0.10$.

RESULTS

Descriptive Data

Table 1 summarizes the key characteristics of each participating dairy herd. All herds were large-scale commercial dairies, predominantly composed of Holstein cows, and located in California ($n = 2$) and Wisconsin ($n = 2$). Most primiparous cows were inseminated with either sexed or conventional semen from dairy breeds, whereas multiparous cows were mainly bred using beef breeds' semen (Angus, Charolais, Simmental, or Limousin). A total of 6,621 cows (primiparous and multiparous) met the inclusion criteria and were fully protocol-compliant, having received all 7 hormonal treatments at the specified time points before insemination: GnRH at d –27, –20, and –17; PGF2 α at d –10, –3, and –2; and GnRH at d –1.

California herds began synchronization at ~ 61 DIM, with AI beginning around 88 DIM, whereas Wisconsin herds initiated synchronization at 43 DIM and insemination around 70 DIM. Pregnancy diagnosis was performed by transrectal ultrasonography at 3 herds and by palpation at 1 (Table 1).

Effects of Treatment on Pregnancy Outcomes

Table 2 presents pregnancy outcomes by treatment and parity. There was no significant treatment-by-parity interaction for first-service P/AI at 32 to 39 d post-AI ($P = 0.13$) or at 66 to 67 d post-AI ($P = 0.14$), allowing comparison of the main effects. Cows treated with 200 μg GnRH had higher first-service P/AI than cows receiving 100 μg (51.8% vs. 48.6%; $P = 0.04$) at 32 to 39 d post-AI and at 66 to 67 post-AI (47.9% vs. 44.8%; $P = 0.02$). Primiparous cows had higher P/AI than multiparous cows overall at 32 to 39 post-AI ($P = 0.02$) and at 66 to 67 post-AI ($P = 0.01$). Although the interaction was not significant, treatment effects varied by parity: among primiparous cows, P/AI was similar between doses at d 32 to 39 ($P = 0.66$) and 66 to 67 ($P = 0.67$) post-AI, whereas among multiparous cows, P/AI was significantly greater in the 200 μg group at d 32 to 39 post-AI (50.7% vs. 45.4%; $P < 0.01$) and d 66 to 67 post-AI (46.6% vs. 41.6%; $P < 0.01$). There were no differences in pregnancy loss between treatments (Table 2).

Culling and death rates did not differ by treatment ($P = 0.24$), and no treatment-by-parity interaction was observed ($P > 0.99$). Primiparous cows had lower removal rates (6.3%) than multiparous cows (10.6%; $P < 0.01$). Average parity among multiparous cows was similar between treatments (3.02 for 100 μg ; 3.00 for 200 μg). Days in milk at AI did not differ by treatment and averaged ~80 d for primiparous and ~78 d for multiparous cows. Analysis of herd-by-treatment interaction for P/AI revealed a significant herd effect ($P = 0.03$) but no interaction with treatment ($P = 0.24$). Thus, the mixed model

results were considered robust across herds. Across herds A through D, the average DIM ranged from 200 to 278 d.

Table 3 presents additional demographics and pregnancy-related outcomes in cows enrolled in the study. The number of inseminations per cow was not significantly affected by treatment overall but was influenced by a significant treatment-by-parity interaction ($P = 0.04$). Among primiparous cows, treatment had no effect (mean = 1.96), but among multiparous cows, those receiving 200 μg required fewer inseminations than those receiving 100 μg (2.08 vs. 2.20; $P = 0.01$). Although overall cumulative P/AI beyond first service did not differ by treatment ($P = 0.92$), it differed by parity (83% in primiparous vs. 77% in multiparous cows; $P = 0.03$). Average days open among pregnant cows were not influenced by treatment ($P = 0.77$) or parity ($P = 0.94$), and no treatment-by-parity interaction was observed ($P = 0.80$). Early embryonic loss, determined from discrepancies between the initial and follow-up pregnancy checks, was not affected by treatment or parity ($P \geq 0.44$) and averaged 9.6%.

Analysis of herd-by-treatment interaction for P/AI revealed a significant herd effect ($P = 0.03$) but no interaction with treatment ($P = 0.24$). Thus, the mixed model results were considered robust across herds. Table 4 illustrates herd-level variability in P/AI, with consistently greater treatment effects in multiparous cows than in primiparous cows.

Figure 2 illustrates the results stratified by semen type. For conventional semen, there were effects of treatment ($P = 0.002$) and parity ($P < 0.001$), with no interaction between treatment and parity ($P = 0.40$) observed (Fig-

Table 2. First-service pregnancy per AI according to protocol compliance in each herd used for the study¹ (pregnancy per AI is presented as LSM $\pm$ SEM)

Parity group	Primiparous		Multiparous		Totals	Treatment (T) <i>P</i> -value	Parity (P) <i>P</i> -value	T $\times$ P <i>P</i> -value
	100 μg	200 μg	100 μg	200 μg				
G1 dose								
Protocol-compliant cows, n	1,235	1,290	2,075	2,021	6,621	—	—	—
Herd A	442	446	780	792	2,460	—	—	—
Herd B	149	183	316	300	948	—	—	—
Herd C	275	270	359	341	1,245	—	—	—
Herd D	369	391	620	588	1,968	—	—	—
Cows pregnant at first pregnancy check, n	637	678	947	1,026	3,288	—	—	—
Herd A	231	224	364	386	1,205	—	—	—
Herd B	75	88	158	166	487	—	—	—
Herd C	132	133	148	174	587	—	—	—
Herd D	199	233	277	300	1,009	—	—	—
Pregnancy per first AI, ¹ % (d 32–39 post-AI)	51.8 $\pm$ 1.7	52.8 $\pm$ 1.6	45.4 $\pm$ 1.4	50.7 $\pm$ 1.4	—	0.04	0.02	0.13
Pregnancy per first AI, ¹ % (d 66–67 post-AI)	48.0 $\pm$ 1.4	49.2 $\pm$ 1.4	41.6 $\pm$ 1.1	46.6 $\pm$ 1.1	—	0.02	0.01	0.14
Pregnancy loss, from 32–39 to 66–67 post-AI	8.6 $\pm$ 1.1	8.1 $\pm$ 1.1	10.7 $\pm$ 0.9	9.9 $\pm$ 0.9	—	0.52	0.06	0.90
Percent difference from 2 mL dose, d 32–39	1.9		11.7		—	—	—	—
Cull or dead cows, % of protocol-compliant	6.7 $\pm$ 0.9	5.9 $\pm$ 0.8	11.2 $\pm$ 1.1	10.0 $\pm$ 1.0	—	0.24	<0.01	>0.99

¹Data were analyzed by GLIMMIX to assess fixed effects of treatment, parity, treatment $\times$ parity, and random effects of herd, service sire, herd $\times$ treatment $\times$ parity, month fresh within parity $\times$ herd. Reported values are LSM $\pm$ SEM. With the overall effect of treatment significant at $P = 0.04$, within primiparous cows, the comparison of 2 to 4 mL was not significant ($P = 0.66$), although within multiparous cows, the comparison of 2 to 4 mL was significant ($P = 0.009$).

Table 3. Demographic, reproductive, and performance data from the study¹ (LSM $\pm$ SEM)

Parity group	Primiparous		Multiparous		Treatment (T) <i>P</i> -value	Parity (P) <i>P</i> -value	T $\times$ P <i>P</i> -value
G1 dose ²	100 μ g	200 μ g	100 μ g	200 μ g			
Average parity	1.00	1.00	3.02	3.00	—	—	—
Days in milk at first insemination	79.7 $\pm$ 5.5	79.8 $\pm$ 5.5	77.7 $\pm$ 5.5	77.8 $\pm$ 5.5	0.99	0.72	0.99
Average inseminations, ³ n	1.94 $\pm$ 0.05	1.97 $\pm$ 0.05	2.20 $\pm$ 0.04	2.08 $\pm$ 0.04	0.25	<0.01	0.04
Total pregnancies, ³ n	1,035	1,061	1,581	1,583	—	—	—
Total P/AI, ³ %	83.6 $\pm$ 1.6	81.7 $\pm$ 1.6	76.4 $\pm$ 1.7	78.4 $\pm$ 1.6	0.92	0.03	0.11
No. early embryonic loss, ⁴ n	56	56	104	104	—	—	—
Early embryonic loss, ⁴ %	9.1 $\pm$ 1.8	8.6 $\pm$ 1.7	10.7 $\pm$ 1.8	10.1 $\pm$ 1.7	0.66	0.44	0.99

¹Data were analyzed by PROC MIXED or PROC GLIMMIX to assess fixed effects of treatment, parity, treatment $\times$ parity, and random effects of herd, service sire, herd $\times$ treatment $\times$ parity, month fresh within parity $\times$ herd. Reported values are LSM $\pm$ SEM.

²First GnRH dose of the breeding Ovsynch, given 10 d before insemination.

³Based on data through the last data pull date at each herd: herd A, June 21, 2021; herd B, September 20, 2021; herd C, September 13, 2021; herd D, October 2, 2021. Average DIM at the last data pull for herds A, B, C, and D were 200 $\pm$ 44, 200 $\pm$ 44, 200 $\pm$ 44, and 200 $\pm$ 44, respectively (mean $\pm$ SD).

⁴Cows that were pregnant at initial postinsemination pregnancy checks (32–39 d) but were open on recheck at 66 to 67 d postinsemination.

ure 2A), resembling the overall responses for the study. However, the sexed semen subset revealed no effects of treatment, parity, or an interaction between treatment and parity (Figure 2B).

Effects of Treatment on Ovarian Parameters and Progesterone Concentration

Cows receiving 200 μ g GnRH at G1 had greater ovulatory response compared with cows receiving 100 μ g ($P = 0.03$; Figure 3A). Ovulation was also higher in multiparous than in primiparous cows ($P = 0.03$; Figure 3B), with no treatment-by-parity interaction ($P = 0.61$). When the P4 category (Low <3.0 ng/mL; High $\geq$ 3.0 ng/mL) was included in the model, a significant 3-way interaction was detected ($P = 0.04$): primiparous cows with Low-P4 showed improved ovulatory response with 200 μ g treatment when compared with 100 μ g, while, for multiparous cows, High-P4 showed improved ovulatory response with 200 μ g treatment with 200 μ g (Figure 4).

Double ovulation rates were higher in multiparous than primiparous cows (17.7% vs. 8.8%; $P = 0.02$) and

tended to be higher in Low-P4 cows than High-P4 cows (16.0% vs. 9.4%; $P = 0.07$). Treatment had no significant effect on double ovulation rates (100 μ g = 9.7%; 200 μ g = 13.7%).

At G1, the presence of a functional CL (P4 >1.0 ng/mL) was similar between treatments (95.1% for 100 μ g; 95.8% for 200 μ g; $P = 0.56$), with no effect of parity or interaction ($P > 0.60$; Table 5). The P4 concentrations at G1 also did not differ (100 μ g = 4.7 $\pm$ 0.2 ng/mL; 200 μ g = 4.4 $\pm$ 0.2 ng/mL; $P = 0.45$).

Other ovarian parameters at G1, including CL number, luteal volume, follicle $\geq$ 10 mm presence, and follicular volume, were not different between treatments or treatment-by-parity groups. However, multiparous cows exhibited greater ($P < 0.01$) values for several parameters (multiple CL number, mean CL number, luteal volume, multiple follicle presence, the diameter of the largest follicle, mean follicle number, and mean follicular volume; Table 5).

Similar patterns were observed at the time of PGF_{2 α} administration (Table 6). Treatment did not affect P4 concentrations (100 μ g = 3.6 $\pm$ 0.2 ng/mL; 200 μ g = 3.4

Table 4. Herd-specific study results related to first-service conception risk (number of cows represented) following treatment with 100 μ g/2 mL or 200 μ g/4 mL of gonadorelin hydrochloride at the first GnRH dose of the breeding Ovsynch (G1) in the study¹

Parity group	Primiparous first-service P/AI			Multiparous first-service P/AI		
G1 dose ² (herd)	100 μ g	200 μ g	Treatment difference 200–100 μ g	100 μ g	200 μ g	Treatment difference 200–100 μ g
A	52.2 (442)	50.1 (446)	–2.1	46.5 (780)	48.6 (792)	2.1
B	52.6 (149)	50.4 (183)	–2.2	50.3 (316)	56.2 (300)	5.9
C	48.2 (275)	49.5 (270)	1.3	41.2 (359)	51.0 (341)	9.8
D	53.8 (369)	59.4 (391)	5.6	44.5 (620)	50.9 (588)	6.4

¹All cows were enrolled in a Double-Ovsynch estrus synchronization protocol; data are based on pregnancy outcomes evaluated 32 to 39 d following the first postpartum insemination.

²First GnRH dose of the breeding Ovsynch, given 10 d before insemination.

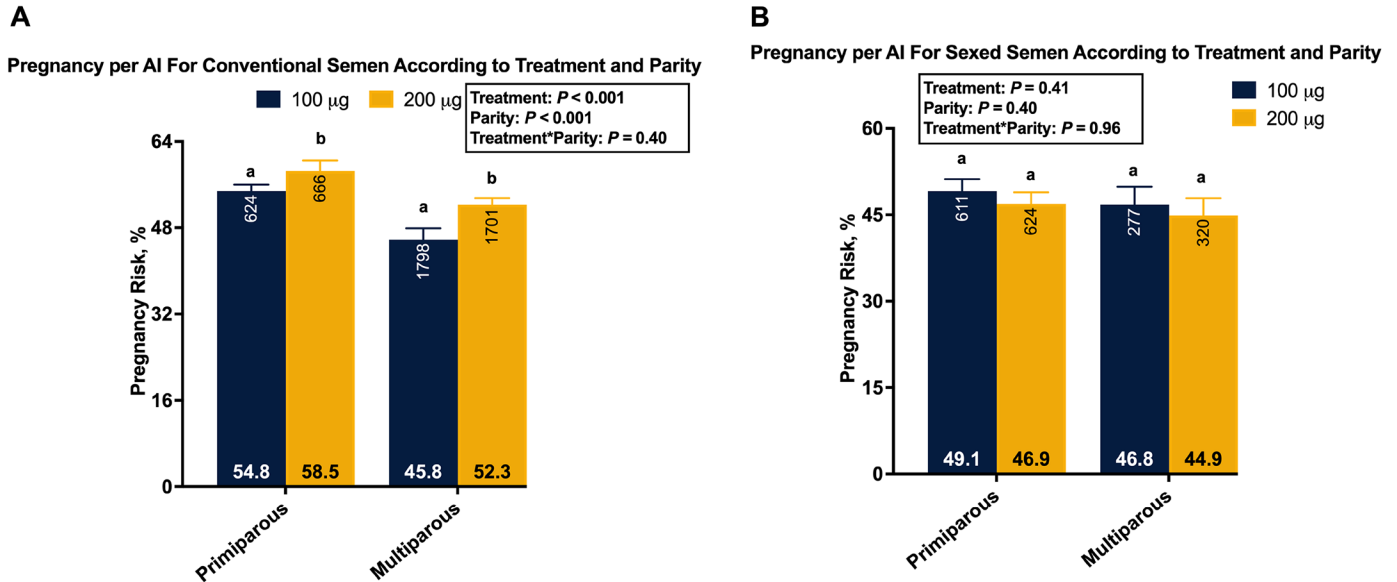


Figure 2. Pregnancy risk according to treatment-by-parity for the subset of cows inseminated using conventional (A) semen (n = 4,789) in herds A, B, and C, and D; and sexed (B) semen (n = 1,832) in herds A, B, C, and D of study cows evaluated in the study.

± 0.2 ng/mL; $P = 0.58$), but parity had a significant effect, with higher P4 levels in multiparous cows (4.1 ng/mL) compared with primiparous cows (3.0 ng/mL; $P < 0.01$). Ovarian structure measurements at PGF_{2α} followed similar trends as at G1, with higher values ($P < 0.03$) ob-

served in multiparous cows for the following parameters (multiple CL number, mean CL number, luteal volume, multiple follicle presence, the diameter of the largest follicle, mean follicle number, and mean follicular volume).

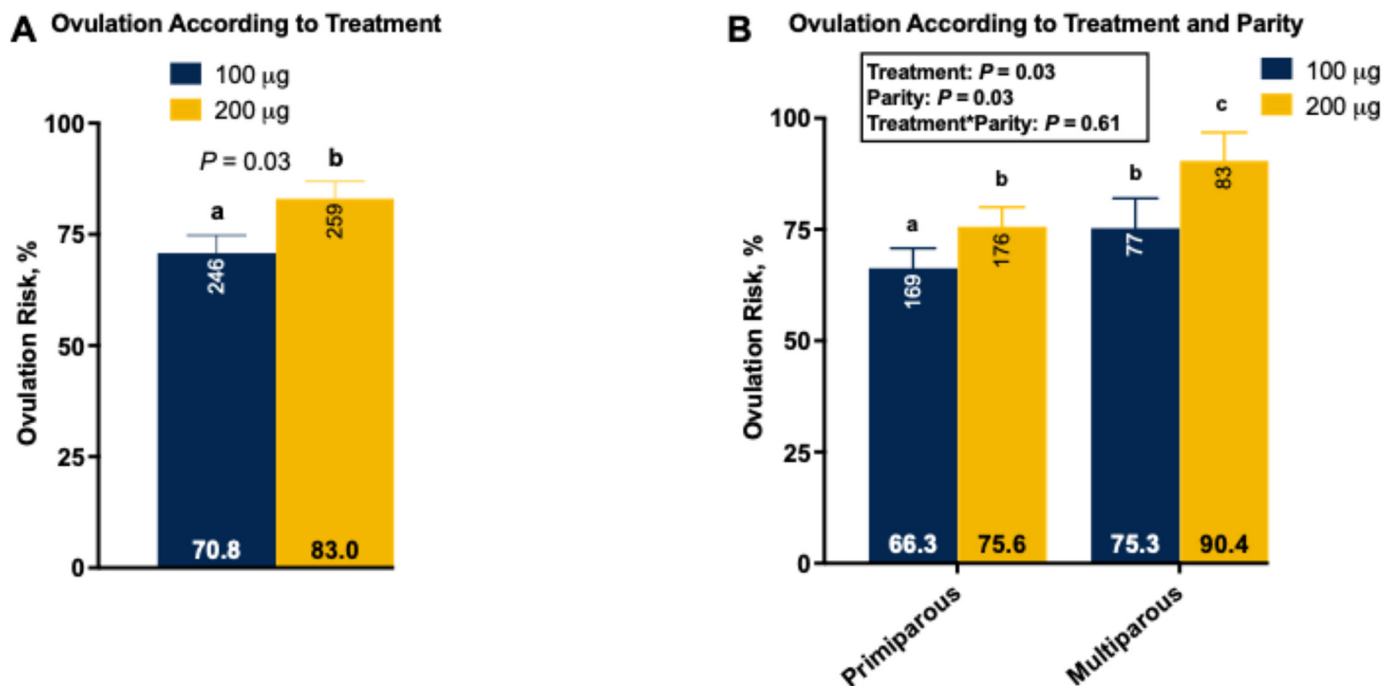


Figure 3. Ovulation risk according to treatment (A) and ovulation risk according to treatment-by-parity for the subset (n = 505) of study cows evaluated in herd A (B). Values within the top of the bar represent the number of cows in each category. Error bars represent the SEM.

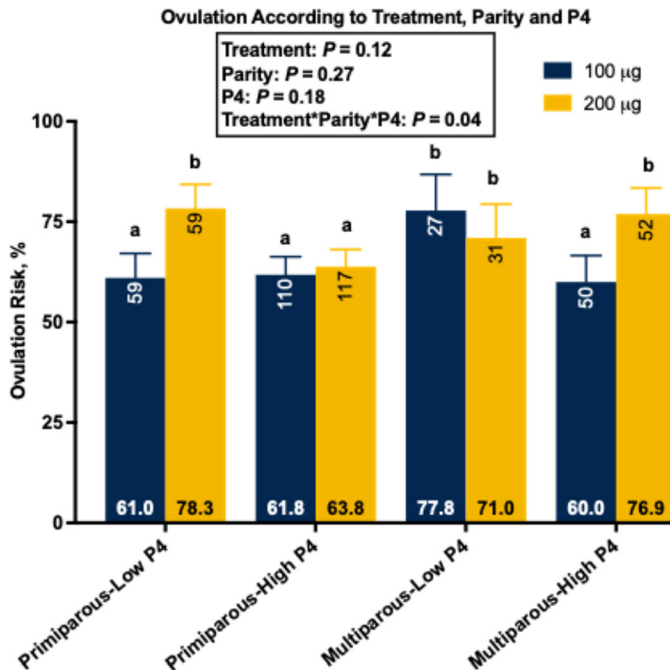


Figure 4. Ovulation risks by parity groups, progesterone (P4) classification categories, and treatments for the subset of study cows evaluated in herd A. Values within parity-by-treatment bars represent the number of cows with progesterone concentrations classified as low (<3 ng/mL) or high (≥3.00 ng/mL) in blood samples collected at the first injection of GnRH of the breeding Ovsynch (G1). Error bars represent the SEM.

Effect of Treatment on LH Secretion

Cows administered 200 µg of GnRH had significantly higher LH peak concentrations ($P = 0.04$) and greater LH AUC ($P < 0.001$) than those administered 100 µg (Table

7). Time to LH peak did not differ ($P = 0.18$). Mean LH concentrations throughout the sampling period varied over time ($P < 0.001$) and were higher in cows treated with 200 µg ($P < 0.001$), but there was no treatment-by-time interaction ($P = 0.23$; Figure 5).

DISCUSSION

The findings of this large, multiherd randomized trial support the hypothesis that increasing the dose of gonadorelin hydrochloride from 100 to 200 µg at the first GnRH of the breeding Ovsynch in a DO program improves reproductive performance in lactating dairy cows. This effect was evidenced by increased P/AI at first service, enhanced ovulatory response, and greater LH release in cows treated with the higher dose. Notably, the magnitude of the response varied with parity, suggesting nuanced physiological interactions among dose, hormonal milieu, and lactational status.

The DO protocol (Souza et al., 2008) is widely adopted due to its superior fertility outcomes and tighter physiological regulation compared with other protocols such as Presynch-Ovsynch (Moreira et al., 2001) and G6G (Bello et al., 2006). Indeed, in addition to placing most cows out of anovulatory status (Souza et al., 2008; Herlihy et al., 2012; Carvalho et al., 2018). Indeed, benefits attributed to DO include the creation of an optimum environment for follicle development and maintenance of oocyte quality that blocks prolonged exposure to high LH pulses that presumptively prevents early meiotic resumption, reduces double ovulation, twin pregnancies, and pregnancy loss, and leads to ovulation of smaller follicles similar to heifers thought to be more fertile at timed AI

Table 5. Ovarian structures and serum progesterone (P4) concentrations immediately before G1¹ of Double-Ovsynch for the subset of 505 cows²

Parity group	Primiparous		Multiparous		Treatment (T) P-value	Parity (P) P-value	T × P P-value
	100 µg	200 µg	100 µg	200 µg			
G1 dose ¹							
CL present, ³ % (n/n)	92.9 (157/169)	85.6 (151/176)	92.2 (71/77)	90.3 (75/83)	0.14	0.61	0.64
Functional CL, ⁴ (P4 >1.0 ng/mL, n)	95.9 (162/169)	95.4 (168/176)	93.5(72/77)	93.6 (80/83)	0.56	0.43	0.80
Multiple CL, % (n/n)	11.8 (20/169)	14.6 (20/176)	22.1 (17/77)	22.9 (19/83)	0.60	<0.01	0.74
Mean CL number, n	1.04 ± 0.04	1.02 ± 0.04	1.16 ± 0.06	1.18 ± 0.06	0.90	<0.01	0.73
Luteal volume, ⁴ mM	11,203 ± 458	11,212 ± 467	13,242 ± 686	13,300 ± 654	0.95	<0.01	0.96
Progesterone, ng/mL	4.3 ± 0.3	4.6 ± 0.3	5.1 ± 0.4	4.3 ± 0.4	0.45	0.39	0.11
Follicles ≥10 mm present, % (n/n)	98.8 (167/169)	97.7 (172/176)	100.0 (77/77)	97.6 (81/83)	0.07	0.27	0.25
Multiple follicle present, % (n/n)	48.5 (82/169)	63.1 (111/176)	67.5 (52/77)	71.1 (59/83)	0.60	<0.01	0.74
Largest follicle diameter, ⁴ mm	16.60 ± 0.38	16.89 ± 0.38	18.26 ± 0.53	17.83 ± 0.54	0.88	<0.01	0.45
Mean follicle number, n	1.57 ± 0.06	1.79 ± 0.06	1.94 ± 0.09	2.04 ± 0.08	0.04	<0.01	0.43
Mean follicular volume, ⁴ mM	3,383 ± 408	3,862 ± 402	5,816 ± 602	5,252 ± 586	0.93	<0.01	0.31

¹First GnRH dose of the breeding Ovsynch of DO, given 10 d before insemination.

²Results are presented as either percentage [(number of cows with specific response/total number of cows available to evaluate the response) × 100%] or mean ± SEM.

³Defined based on the presence of luteal structures assessed by transrectal ultrasound.

⁴Defined based on serum progesterone concentrations >1.0 ng/mL.

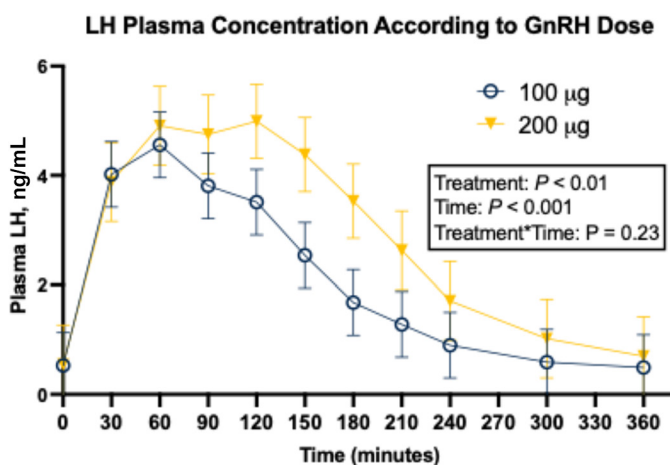
Table 6. Ovarian structures and serum progesterone (P4) concentrations immediately before the first PGF_{2α}¹ of Double-Ovsynch for cows in study 1A²

Parity group	Primiparous		Multiparous		Treatment (T) P-value	Parity (P) P-value	T × P P-value
G1 dose ³	100 µg	200 µg	100 µg	200 µg			
CL present, ⁴ % (n/n)	97.0 (164/169)	95.4 (168/176)	100.0 (77/77)	98.8 (82/83)	0.19	0.01	0.34
Functional CL, ⁵ (P4 >1.0 ng/mL, n)	97.6 (165/169)	96.6 (170/176)	92.2 (71/77)	97.6 (81/83)	0.41	0.41	0.11
Multiple CL, % (n/n)	66.2 (112/169)	71.6 (126/176)	74.0 (57/77)	81.9 (68/83)	0.11	0.03	0.63
Mean CL number, n	1.69 ± 0.05	1.76 ± 0.07	1.90 ± 0.07	2.08 ± 0.07	0.06	<0.01	0.41
Luteal volume, ⁴ mM	15,600 ± 603	15,106 ± 597	21,344 ± 881	20,280 ± 854	0.29	<0.01	0.70
Progesterone, ng/mL	3.1 ± 0.3	2.9 ± 0.2	4.1 ± 0.4	4.0 ± 0.4	0.58	<0.01	0.99
Follicles ≥10 mm present, % (n/n)	100 (169/169)	100 (176/176)	100.0 (77/77)	100.0 (83/83)	—	—	—
Multiple follicles present, % (n/n)	59.2 (100/169)	59.7 (105/176)	74.0 (57/77)	67.5 (56/83)	0.47	0.01	0.41
Largest follicle diameter, mm	15.75 ± 0.37	15.25 ± 0.37	17.56 ± 0.55	17.18 ± 0.53	0.35	<0.01	0.90
Mean follicle number, n	1.71 ± 0.06	1.79 ± 0.06	2.10 ± 0.08	1.82 ± 0.08	0.16	<0.01	0.01
Mean follicular volume, ⁴ mM	3,191 ± 350	3,094 ± 344	5,522 ± 520	4,442 ± 501	0.98	<0.01	0.26

¹First injection of PGF_{2α}, given 3 d before insemination.²Results are presented as either percentage [(number of cows with specific response/total number of cows available to evaluate the response) × 100%] or mean ± SEM.³First GnRH dose of the breeding Ovsynch, given 10 d before insemination.⁴Defined based on the presence of luteal structures assessed by transrectal ultrasound.⁵Defined based on serum progesterone concentrations >1.0 ng/mL.

(Giordano et al., 2013; Carvalho et al., 2018; Martins et al., 2018; Fricke and Wiltbank, 2022). The conundrum that was the primary motive for the current study is that, despite these benefits, ovulation in response to the first GnRH of the breeding Ovsynch remained suboptimal and, in some studies, not higher than that of cows presynchronized with PGF_{2α} (Souza et al., 2008; Herlihy et al., 2012). Prior work suggests that increasing GnRH dose at G1 can overcome P4-suppressed LH release (Giordano et al., 2012) and the presumed effect on ovulation and pregnancy (Valdes-Arciniega et al., 2023).

Gonadorelin hydrochloride (Factrel; 50 µg/mL) is US Food and Drug Administration-approved at doses of 100 to 200 µg for use in synchronization programs (FDA, 2013). The drug mimics endogenous GnRH and exerts its effect by stimulating LH and FSH secretion, thereby promoting ovulation of the dominant follicle and initiation of a new follicular wave (Fricke et al., 1998; Chenault et al., 2014). In a multiherd field study, Chenault et al. (2014) reported numerically higher P/AI rates with 200 µg compared with 100 µg, though differences were not significant due to study design limitations. In the present study, cows treated with 200 µg GnRH at G1 had significantly higher first-service P/AI than cows receiving 100 µg (51.8% vs. 48.6%), and this effect was primarily driven by responses in multiparous cows (50.7% vs. 45.4%). In primiparous cows, no significant treatment effect was observed (52.8% vs. 51.8%). These findings reinforce the importance of considering parity in synchronization protocols, as primiparous and multiparous cows differ in endocrine profiles, metabolic status, and steroid clearance (Wiltbank et al., 2011) and are aligned with the results from a previous study (Valdes-Arciniega et al., 2023).

**Figure 5.** Mean (±SEM) LH concentrations in lactating dairy cows treated with 100 µg (n = 10) or 200 µg (n = 8) of GnRH as gonadorelin hydrochloride at the first GnRH (G1) of the breeding Ovsynch portion of the Double-Ovsynch protocol. Asterisks denote a difference (Tukey mean separation test) in mean LH concentration between treatment groups.**Table 7.** Effect of 100 µg versus 200 µg of GnRH as gonadorelin hydrochloride at first GnRH of the breeding Ovsynch of the Double-Ovsynch on LH secretion in lactating dairy cows

Item	100 µg (n = 10)	200 µg (n = 8)	P-value
LH peak concentration, ng/mL	4.0 ± 0.93	6.7 ± 0.93	0.04
Time to LH peak in min	70 ± 3.5	77 ± 3.5	0.18
AUC (units)	740 ± 42	1,040 ± 120	0.001

Indeed, Valdes-Arciniega et al. (2023) reported a treatment-by-semen type interaction for ovulation and P/AI in primiparous cows receiving 100 or 200 µg of gonadorelin hydrochloride at G1 with sexed semen. Their data showed that the 200-µg dose significantly improved ovulation and P/AI in multiparous cows, while primiparous cows showed no benefit, which was largely attributed to the effect of semen type and the fact that primiparous cows tend to be inseminated more frequently by sexed semen, which seemed to offset the benefits of a higher dose of GnRH. In our study, sexed semen was not used at all farms and could not be included in the final statistical models. When we analyzed only the 3 herds with data on conventional sexed semen, we were unable to detect an effect of semen type or the interaction between semen type and treatment. However, when we analyzed the data stratified by semen type, we found that results were not significant for treatment, parity, or their interaction for sexed semen, whereas for conventional semen, there were effects of treatment and parity, with no treatment-by-parity interaction. The combined lack of results for sexed semen and the positive results for conventional semen, with a greater effect size than the overall results, align with Valdes-Arciniega et al. (2023), which underscores responses to GnRH dose present only in conventional semen. Yet, when we analyzed with just conventional semen, the magnitude of increase between primiparous versus multiparous (3.7% vs. 6.5%) indicated that the outcomes of the present study support the notion that multiparous cows (likely due to higher P4 metabolism and attenuated LH responsiveness) may require greater stimulation to achieve similar ovulatory success (Colazo et al., 2009; Giordano et al., 2012).

The LH data from our study also showed that cows receiving 200 µg GnRH had significantly higher peak LH concentrations and a greater AUC, indicating enhanced pituitary responsiveness. These findings are consistent with those of Souza et al. (2009), who reported dose-dependent increases in LH secretion in response to 50 µg versus 100 µg GnRH. Giordano et al. (2012) extended this work by demonstrating that 200 µg GnRH can enhance LH release even in the presence of high circulating P4, which did not seem to be as pronounced in cows receiving 100 µg of GnRH. An interesting finding of our study was a 3-way interaction between treatment, parity, and P4 category (Low <3.0 ng/mL, High ≥3.0 ng/mL). Specifically, primiparous cows with low P4 and multiparous cows with high P4 exhibited greater ovulatory response when treated with 200 µg. Giordano et al. (2012) also noted that elevated P4 attenuated LH release and ovulatory response when 100 µg was used but that 200 µg could overcome this inhibition. In our study, the high overall proportion of cows (~67%) with P4 ≥3.0 ng/mL at G1 underscores the relevance of this hormonal inter-

action. The disparity between parity groups may reflect differing metabolic profiles caused by greater milk production in multiparous cows, which may lead to greater P4 clearance and follicular sensitivity (Sangsritavong et al., 2002; Sartori et al., 2004). Therefore, it is reasonable to surmise that multiparous cows require higher LH surges to achieve ovulation under elevated P4 conditions. Double ovulation rates were higher in multiparous than in primiparous cows and tended to be higher in low P4 cows, which might help explain the more pronounced benefit observed in multiparous cows. This finding supports the hypothesis that the endocrine environment at G1 influences follicular recruitment and ovulation rate, which may ultimately affect luteal function and fertility (Cerri et al., 2009). Progesterone concentrations were measured using an ELISA with a limit of detection of 1.17 ng/mL and relatively high intra-assay CV. Although the assay is validated and widely used for bovine samples, these analytical characteristics may reduce precision near the lower physiological range. Accordingly, P4-based categorizations and 3-way interactions in relatively small samples should be interpreted cautiously, with emphasis placed on robust treatment and parity effects that were consistent across analyses. Importantly, conclusions regarding ovulatory response and P/AI do not depend on absolute P4 values and we acknowledge the potential risk of type-1 error associated with our 3-way interaction.

From a practical standpoint, the data suggest that benefits are more likely to be achieved and replicated in multiparous cows receiving conventional semen followed by primiparous cows receiving conventional semen, making this a potential strategy for farms capable of strategically treating these cows without compromising compliance. In contrast, considering that the benefits observed might be linked to milk production and semen type, not parity itself, and the challenges of implementing different treatments across parity groups or semen types in the same herd, implementing a strategy of only increasing the dose of GnRH can risk limiting the potential benefits of a higher dose at the beginning of the Double-Ovsynch. The current study is also consistent with earlier observations that ovulation to the first GnRH injection improves synchronization and subsequent fertility (Galvão and Santos, 2010). By increasing the ovulatory response at G1, the second GnRH in the breeding Ovsynch more reliably induces ovulation of a synchronized, mature follicle, thereby improving the likelihood of pregnancy. Some of the herd differences observed in the study may be attributed to variations in diet and milk production, which could influence P4 clearance and the potential benefits associated with a high GnRH dose. For instance, in the current study, the 2 farms with greater benefits from the higher GnRH dose are those with the highest milk production per cow, a factor that affects the odds

of double ovulation and the frequency of cows with multiple CL and high progesterone. Moreover, the difference in semen type used in the current study might also help explain the observed differences.

Importantly, the study by Valdes-Arciniega et al. (2023) adds further nuance. They demonstrated that the improved ovulatory response in multiparous cows treated with 200 µg was associated with greater pregnancy retention through 39 d post-AI. Although our study did not detect differences in early embryonic loss, the direction of effect (higher first-service P/AI and fewer services per conception in multiparous cows) suggests a similar potential benefit. This consistency between studies using different herds and reproductive management systems underscores the robustness of the parity-specific effect.

While early pregnancy loss and days open were not significantly affected, the reduction in services per conception among multiparous cows treated with 200 µg suggests tangible reproductive and economic benefits. Wiltbank et al. (2011) emphasized that each additional insemination imposes both direct and indirect costs, including semen, labor, and lost production time. Therefore, the observed 11.7% relative increase in P/AI in multiparous cows treated with 200 µg represents a potentially cost-effective management intervention, warranting further economic evaluation.

Taken together, these findings highlight the complexity of hormonal responses in synchronization protocols and emphasize the importance of stratifying results by parity and endocrine status. The improved ovulatory and fertility outcomes with the 200-µg dose of GnRH, especially in multiparous cows, support the use of individualized reproductive management strategies tailored to cow physiology.

CONCLUSIONS

This large-scale, randomized study involving more than 6,600 lactating dairy cows across 4 commercial herds demonstrated that increasing the GnRH dose from 100 to 200 µg at the first GnRH of the breeding Ovsynch in a DO protocol improves first-service P/AI, particularly in multiparous cows. These effects were accompanied by enhanced ovulatory responses and LH secretion, suggesting a physiological basis for the improved reproductive outcomes. The 3-way interaction among treatment, P4 concentration, and parity further underscores the importance of hormonal status and cow-level factors in modulating response, and stratified data analysis by semen type suggests that conventional semen may be a factor in optimizing responses to an increased dose of GnRH. Future research should aim to elucidate the underlying mechanisms of these interactions and assess the

cost-effectiveness of higher-dose GnRH protocols across diverse production systems.

NOTES

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Nonstandard abbreviations used: AI = artificial insemination; AUC = area under the curve; CL = corpora lutea; DO = Double-Ovsynch; G1 = first GnRH; High-P4 = >3.0 ng/mL; Low-P4 = ≤3.0 ng/mL; P4 = plasma progesterone; P/AI = per AI.

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